Day 6 Assignment

Overview

The Contact Manager application is a basic TypeScript-based console application designed to manage user contacts efficiently. It demonstrates object-oriented programming concepts using TypeScript, including interfaces, classes, strong typing, and method-based data handling. The application allows users to add, view, modify, and delete contacts while ensuring proper validation and error handling.

Key Insights

- Used a TypeScript Interface to define the structure of a Contact
- Implemented a ContactManager class to manage contact operations
- Applied strong typing for better type safety and code reliability
- Used Partial<Contact> for flexible update operations
- Included proper error handling when contacts do not exist
- Provided confirmation messages for successful operations
- Demonstrated encapsulation using a private contacts array

Code Snippets

Output in local Terminel (Powershell) – Based on Test Class

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS    powershell

PS D:\Wipro\Assignments\Day 6\ContactManager> npx tsc contactManager.ts
PS D:\Wipro\Assignments\Day 6\ContactManager> node contactManager.js
Contact added successfully.
Contact added successfully.
Contact added successfully.
Contact added successfully.
Contact modified successfully.
Contact modified successfully.
Contact deleted successfully.
Final Contacts:
[
  {
    id: 1,
    name: 'Siva',
    email: 'hemanth@gmail.com',
    phone: '9876543210'
  },
  {
    id: 2,
    name: 'roshini',
    email: 'roshini@gmail.com',
    phone: '8312045679'
  },
  {
    id: 3,
    name: 'Satya',
    email: 'satya@gmail.com',
    phone: '7685902134'
  }
]
PS D:\Wipro\Assignments\Day 6\ContactManager>
```

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Testing Class

- Added 4 contacts
- Modified name in contact 1
- Modified name in contact 2
- Deleted contact 4

```javascript
// Testing the class

const manager = new ContactManager();

manager.addContact({
    id: 1,
    name: "Hemanth",
    email: "hemanth@gmail.com",
    phone: "9876543210"
});
manager.addContact({
    id: 2,
    name: "roshini",
    email: "roshini@gmail.com",
    phone: "9123456780"
});
manager.addContact({
    id: 3,
    name: "Satya",
    email: "satya@gmail.com",
    phone: "7685902134"
});
manager.addContact({
    id: 4,
    name: "Venkat",
    email: "venkat@gmail.com",
    phone: "6789012345"
});


manager.modifyContact(1, { name: "Siva" });
manager.modifyContact(2, { phone: "8312045679" });

manager.deleteContact(4);

console.log("Final Contacts:");
console.log(manager.viewContacts());
```

<u>Note:</u> Kindly Go through code files in attached zip folder.

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